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Course code: DSA0607

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Course name: Data handling& visualization

Slot D

LAB EXPERIMENTS

6.

Code:

```
Date <- c("2023-01-01",
          "2023-01-02",
          "2023-01-03",
          "2023-01-04",
          "2023-01-05")

Page_Views <- c(1500,1600,1400,1650,1800)
Click_Through_Rate <- c(2.3,2.7,2.0,2.4,2.6)
website <- data.frame(Date,
                      Page_Views,
                      Click_Through_Rate)

print(website)

plot(website$Page_Views,
     type="o",
     col="blue",
     xaxt="n",
     xlab="Date",
     ylab="Page Views",
     main="Daily Page Views Trend")

axis(1,
     at=1:5,
     labels=website$Date)

legend("topleft",
     legend="Page Views",
     col="blue",
```

```

lty=1,
pch=16)
barplot(website$Click_Through_Rate,
        names.arg=website$Date,
        main="Click Through Rate by Date",
        xlab="Date",
        ylab="CTR (%)",
        col="orange",
        border="black")
Likes <- c(120,150,130,160,180)
Shares <- c(60,70,65,75,80)
Comments <- c(30,35,32,40,45)
interaction <- rbind(Likes,
                     Shares,
                     Comments)
x <- 1:5
colors <- c("skyblue","orange","lightgreen")
plot(x,
     Likes,
     type="n",
     ylim=c(0,max(colSums(interaction))),
     xlab="Date",
     ylab="Interactions",
     xaxt="n",
     main="User Interactions")
axis(1,
     at=1:5,
     labels=Date)
polygon(c(x,rev(x)),
        c(rep(0,5),rev(Likes)),

```

```

col=colors[1],

border=NA)

polygon(c(x,rev(x)),

c(Likes,rev(Likes+Shares)),

col=colors[2],

border=NA)

polygon(c(x,rev(x)),

c(Likes+Shares,

rev(Likes+Shares+Comments)),

col=colors[3],

border=NA)

legend("topleft",

legend=c("Likes","Shares","Comments"),

fill=colors)

```

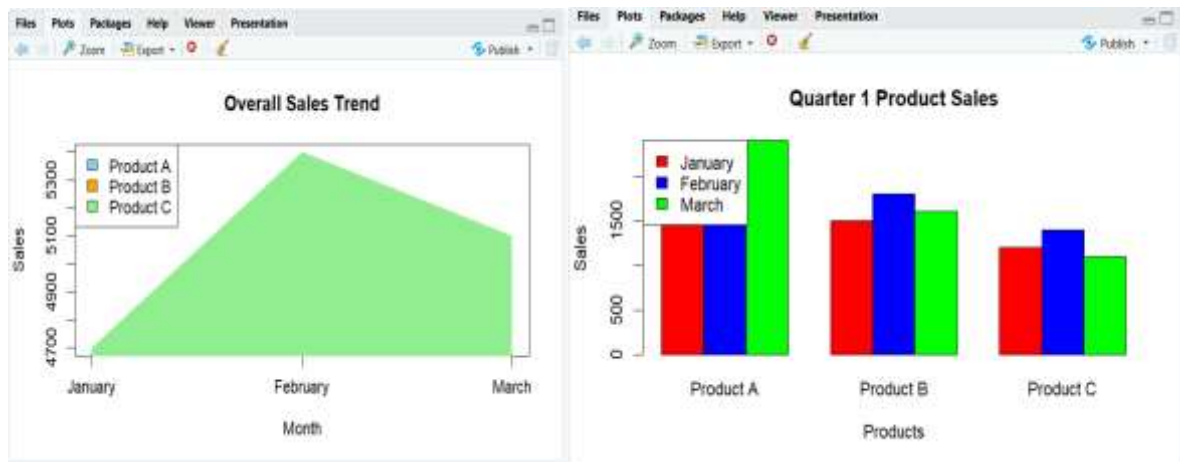
OUTPUT:



```

+ "lightgreen"))
> sales
  Product_ID Product_Name January_Sales February_Sales March_Sales
1          1   Product A          2000           2200          2400
2          2   Product B          1500           1800          1600
3          3   Product C          1200           1400          1100
>

```



7.

Code:

```
# Customer Demographics Dataset
```

```
Customer_ID <- c(1,2,3)
```

```
Age <- c(28,35,42)
```

```
Gender <- c("Female","Male","Female")
```

```
Income <- c(50000,60000,75000)
```

```
customer <- data.frame(Customer_ID,
```

```
    Age,
```

```
    Gender,
```

```
    Income)
```

```
print(customer)
```

```
barplot(customer$Age,
```

```
    names.arg=customer$Customer_ID,
```

```
    main="Customer Age Distribution",
```

```
    xlab="Customer ID",
```

```
    ylab="Age",
```

```
    col="skyblue",
```

```
    border="black")
```

```
gender_count <- table(customer$Gender)
```

```
pie(gender_count,
```

```
    main="Customer Gender Distribution",
```

```
labels=names(gender_count),
```

```
col=c("pink","lightblue"))
```

```
View(customer)
```

OUTPUT:

	Customer_ID	Age	Gender	Income
1	1	28	Female	50000
2	2	35	Male	60000
3	3	42	Female	75000

Environment History Connections Tutorial

R Global Environment

customer 3 obs. of 4 variables

data 6 obs. of 8 variables

Values

Age num [1:3] 28 35 42

Course chr [1:6] "R" "R" "SQL" "R" "R" "SQL"

Customer_ID num [1:3] 1 2 3

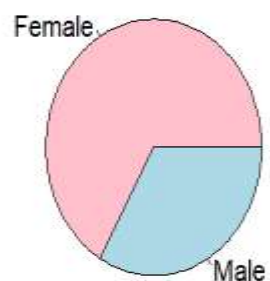
Gender chr [1:3] "Female" "Male" "Female"

gender_count 'table' int [1:2(1d)] 2 1

Files Plots Packages Help Viewer Presentation

Zoom Export Publish

Customer Gender Distribution



8.

Code:

Create Data set

```
Student_ID <- c("L01","L02","L03","L04","L05","L06")
Gender <- c("Male","Female","Male","Female","Male","Female")
Age <- c(20,22,19,21,23,20)
Course <- c("R","R","SQL","R","R","SQL")
Study_Time <- c(3.5,4.2,2.0,5.0,2.5,4.0)
Videos_Watched <- c(12,15,8,18,9,14)
Quiz_Score <- c(78,85,65,92,70,88)
Login_Date <- c("2025-01-05","2025-01-05","2025-02-08",
                "2025-02-08","2025-03-12","2025-03-12")
```

```
data <- data.frame(Student_ID,
                    Gender,
                    Age,
                    Course,
                    Study_Time,
                    Videos_Watched,
                    Quiz_Score,
                    Login_Date)
```

```
print(data)
```

```
hist(data$Quiz_Score,
     main="Distribution of Quiz Scores",
     xlab="Quiz Score",
     ylab="Frequency",
     col="skyblue",
     border="black")
```

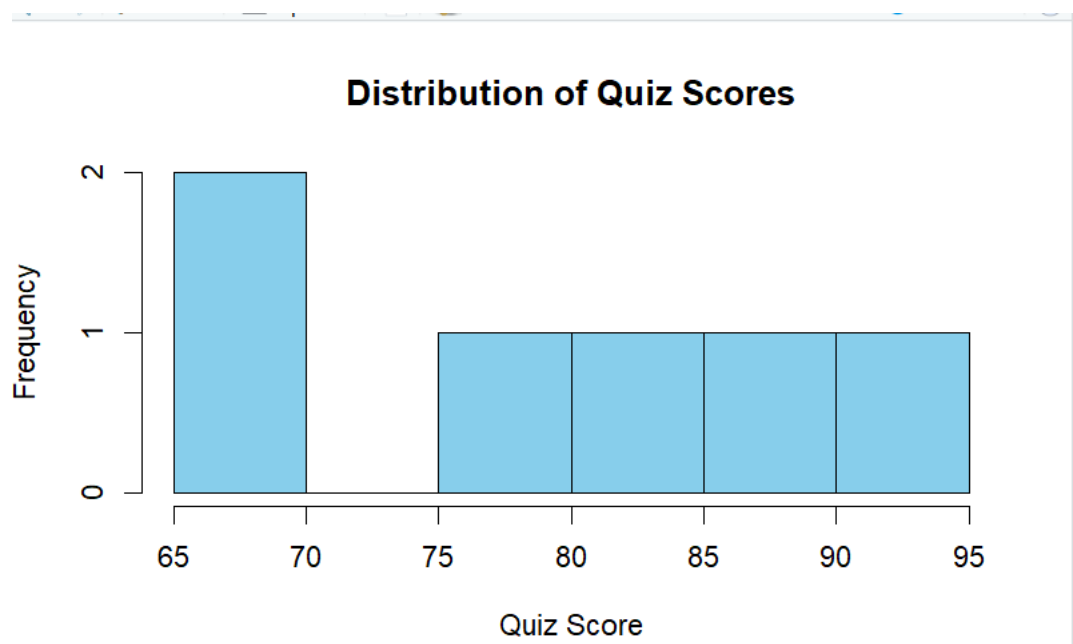
```
plot(avg_score$Quiz_Score,
```

```

type="o",
xaxt="n",
col="blue",
xlab="Month",
ylab="Average Quiz Score",
main="Monthly Average Quiz Score")
axis(1,
     at=1:nrow(avg_score),
     labels=avg_score$Month)
moving_avg <- filter(avg_score$Quiz_Score,
                     rep(1/2,2),
                     sides=1)
lines(moving_avg,
      col="red",
      lwd=2)
legend("topleft",
      legend=c("Average Score","Moving Average"),
      col=c("blue","red"),
      lty=1)

```

OUTPUT:



customer	3 obs. of 4 variables
data	6 obs. of 8 variables
Age	num [1:6] 20 22 19 21 23 20
Course	chr [1:6] "R" "R" "SQL" "R" "R" "SQL"
Customer_ID	num [1:3] 1 2 3
Gender	chr [1:6] "Male" "Female" "Male" "Female" "Male"...
gender_count	'table' int [1:2(1d)] 2 1

9.

Code:

```
# Survey Responses Dataset
```

```
Survey_ID <- c(1,2,3)
```

```
Question1 <- c("A","B","C")
```

```
Question2 <- c("B","A","A")
```

```
Question3 <- c("C","D","B")
```

```
survey <- data.frame(
```

```
  Survey_ID,
```

```
  Question1,
```

```
  Question2,
```

```
  Question3
```

```
)
```

```
print(survey)
```

```
q1 <- table(survey$Question1)
```

```
barplot(q1,
```

```
  main="Distribution of Question 1 Responses",
```

```
  xlab="Responses",
```

```
  ylab="Frequency",
```

```
  col=c("red","blue","green"),
```

```
  border="black")
```

```
response_table <- rbind(
```

```
  table(survey$Question1),
```

```
  table(survey$Question2),
```

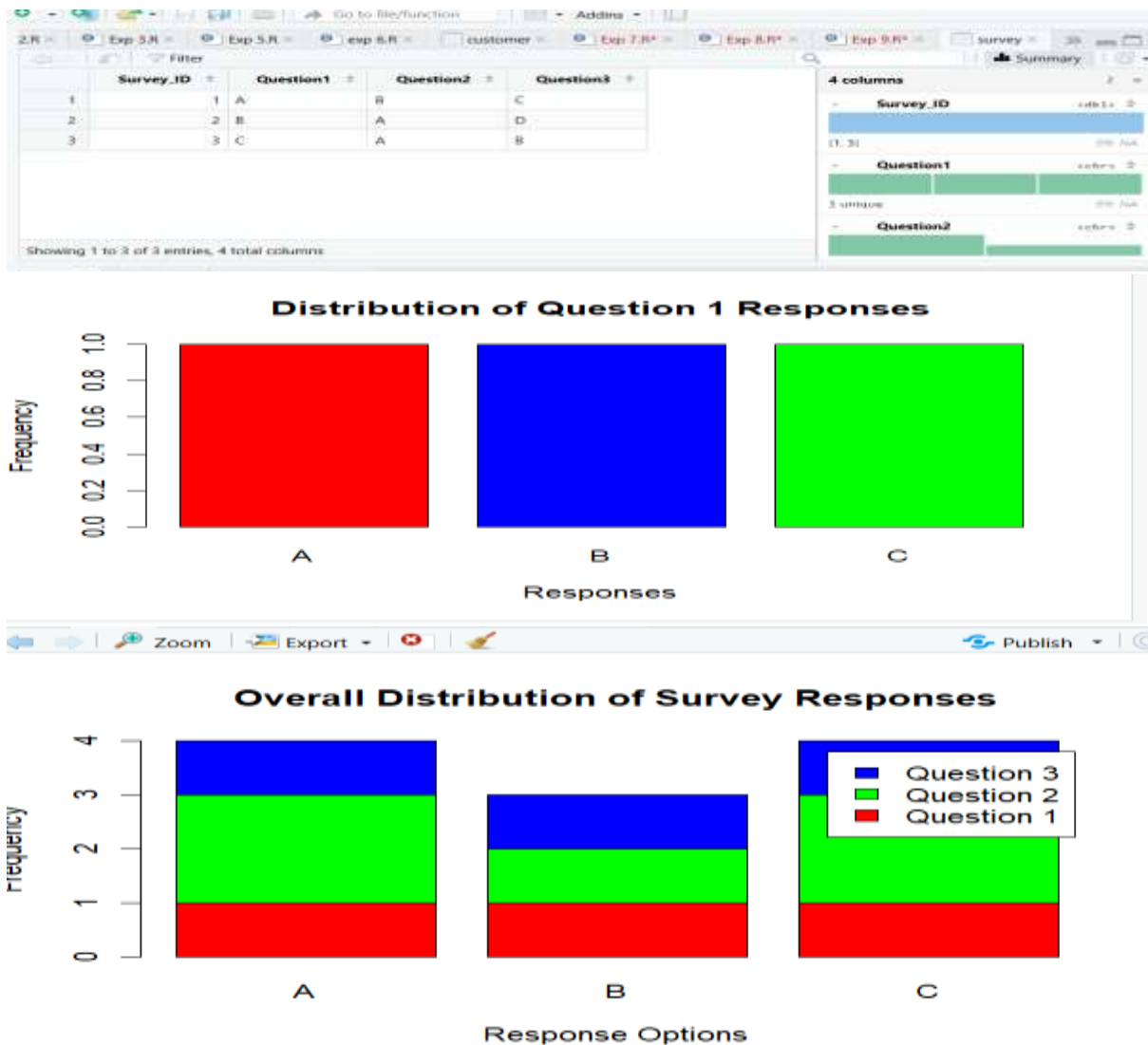


```

table(survey$Question3)
)
barplot(response_table,
        beside=FALSE,
        col=rainbow(nrow(response_table)),
        main="Overall Distribution of Survey Responses",
        xlab="Response Options",
        ylab="Frequency",
        legend.text=c("Question 1","Question 2","Question 3"))
View(survey)

```

OUTPUT:



10.

Code:

```
# Product Category Dataset
Category <- c("Electronics",
              "Clothing",
              "Appliances")
Sales <- c(50000,
           35000,
           40000)
product <- data.frame(Category,
                      Sales)

print(product)
barplot(rev(product$Sales),
        horiz=TRUE,
        names.arg=rev(product$Category),
        col=c("skyblue","orange","lightgreen"),
        main="Sales Conversion Funnel",
        xlab="Sales ($) ",
        border="black")
View(product)
pie(product$Sales,
     labels=product$Category,
     main="Sales Distribution by Product Category",
     col=c("skyblue","orange","lightgreen"))
```

OUTPUT:

